INCH-POUND

MIL-M-28787/434
AMENDMENT 1
3 August 1994

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MEMORY DATA MODULE, KEY CODE JML

This amendment forms a part of Military Specification MIL-M-28787/434, dated 17 May 1991, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGES 4 THROUGH 6

TABLE I: Delete and substitute new table I as printed on pages 2 through 4 of this amendment.

PAGES 55 THROUGH 63

TABLE VI: Delete and substitute new table VI as printed on pages 5 through 13 of this amendment.

PAGE 117

FIGURE 12a: Delete and substitute new figure 12a as printed on page 14 of this amendment.

Custodians:
Army - ER
Navy - NW
Air Force - 85

Review activities:
Army - AT
Navy - MC
Air Force - 13, 19
DLA - ES

Preparing activity:
Navy - NW

(Project 5963-0298)

AMSC N/A FSC 5963
DISTRIBUTION STATEMENT A Approved for public release distribution is unlimited

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:15Z
Check the source to verify that this is the current version before use.

TABLE I Module electrical requirements

Test cond	Parameter 5/	Ref test table	Test type 1/ 3/	25° ± 5°C 4/		Class I 4/ 0° ± 5°C 60° ± 5°C		Class II 4/ -55° ± 5°C 85° ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
A	Isolation	II	S	10	INF	10	INF	10	INF	MOhm
	End-of life	II	S	1	INF	1	INF	1	INF	MOhm
B1 B6	Continuity	III	X	0	1	0	1	0	1	Ohm
C	Vcc stress	IV	Q	--- -- -----Module shell pass all remaining tests----------						
D1 D12	Ibd	V	S	0 0	0 1	0 0	0 1	0.0	0 1	mA
D13,D14	Ibd	V	S	-0 58	0 68	-0 58	0 68	-0.58	0 68	mA
D15	Ibd	V	S	0 0	0 1	0 0	0 1	0.0	0 1	mA
D16	Ibd	V	S	0 0	1 0	0 0	1 0	0.0	1 0	mA
D17	Ibd	V	S	0 0	0 4	0 0	0 4	0.0	0 4	mA
D18	Ibd	V	S	0 0	0 6	0 0	0 6	0 0	0 6	mA
D19,D20	Ibd	V	S	0 0	0 2	0 0	0 2	0.0	0 2	mA
D21	Ibd	V	S	0 0	0 3	0 0	0 3	0 0	0 3	mA
D22	Ibd	V	S	0 0	0 2	0 0	0 2	0 0	0 2	mA
D23	Ibd	V	S	-0 58	0 68	0 58	0 68	-0 58	0 68	mA
D24-D27	Ibd	V	S	0 0	0 1	0 0	0 1	0.0	0 1	mA
D28	Ibd	V	S	-0 58	0 68	0 58	0 68	-0.58	0 68	mA
D29-D32	Ibd	V	S	0 0	0 1	0 0	0 1	0 0	0 1	mA
D33	Ibd	V	S	0 0	0 5	0 0	0 5	0 0	0 5	mA
D34 D37	Ibd	V	S	0 0	0 1	0 0	0 1	0 0	0 1	mA
D38,D39	Ibd	V	S	0 0	0 2	0 0	0 2	0.0	0 2	mA
D40	Ibd	V	S	0 0	0 3	0 0	0 3	0.0	0 3	mA
D41	Ibd	V	S	0 0	0 2	0 0	0 2	0.0	0 2	mA
D42	Ibd	V	S	0 0	0 5	0 0	0 5	0 0	0 5	mA
D43	Ibd	V	S	0 0	0 1	0 0	0 1	0 0	0 1	mA
D44	Ibd	V	S	0 0	0 6	0 0	0 6	0 0	0 6	mA
E1 E18	Ios	VI	S	-350 0	-60 0	350 0	-60 0	-350 0	-60 0	mA
E19 E22	Ios	VI	S	-150 0	-60 0	-150 0	-60 0	-150 0	-60 0	mA
E23	Ios	VI	S	130 0	20 0	-130 0	-20 0	-130 0	-20 0	mA
E24 E27	Ios	VI	S	150 0	60 0	150 0	-60 0	-150 0	-60 0	mA
E28	Ios	VI	S	-130 0	-20 0	-130 0	-20 0	-130 0	20 0	mA
E29 E34	Ios	VI	S	-350 0	-60 0	-350 0	-60 0	-350 0	-60 0	mA

See footnotes at end of table

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:15Z
Check the source to verify that this is the current version before use.

TABLE I Module electrical requirements - Continued

Test cond	Parameter 5/	Ref test table	Test type 1/ - 3/	25° ± 5°C 4/		Class I 4/ 0° ± 5°C 60° ± 5°C		Class II 4/ -55° ± 5°C 85° ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
E35 E38	Ios	VI	S	-150 0	-80 0	-150 0	-60 0	-150 0	-60 0	mA
E39	Ios	VI	S	-130 0	-20 0	-130 0	-20 0	-130 0	-20 0	mA
E40 E43	Ios	VI	S	-150 0	-80 0	-150 0	-60 0	-150 0	-60 0	mA
E44	Ios	VI	S	-130 0	-20 0	-130 0	-20 0	-130.0	-20 0	mA
E45 E52	Ios	VI	S	-350 0	-80 0	-350 0	-60 0	-130 0	-80 0	mA
E53,E54	Ios	VI	S	150 0	-60 0	-150 0	-60 0	-150 0	-60 0	mA
E55	Ios	VI	S	-130 0	-20 0	-130 0	-20 0	-130 0	-20.0	mA
E56 E57	Ios	VI	S	-150 0	-60 0	-150 0	-60 0	-150 0	-60 0	mA
E58	Ios	VI	S	-130 0	-20 0	-130 0	-20 0	-130 0	-20 0	mA
E59 E62	Ios	VI	S	150 0	-60 0	-150 0	-60 0	-150 0	-60 0	mA
E63	Ios	VI	S	-130 0	-20 0	-130 0	-20 0	-130 0	-20 0	mA
E64,E65	Ios	VI	S	-150 0	-60 0	-150 0	-60 0	-150.0	-60 0	mA
E66	Ios	VI	S	-130 0	-20 0	-130 0	-20 0	-130.0	-20 0	mA
E67,E68	Ios	VI	S	-150 0	-60 0	-150 0	-60 0	-150.0	-60 0	mA
E69	Ios	VI	S	-130 0	-20 0	-130 0	-20 0	-130 0	-20 0	mA
E70 E73	Ios	VI	S	-150 0	-60 0	-150 0	-60 0	-150.0	-60 0	mA
E74	Ios	VI	S	-130 0	-20 0	-130 0	-20 0	-130 0	-20 0	mA
E75 E77	Ios	VI	S	-150 0	-60 0	-150 0	-60.0	-150.0	-60 0	mA
E78	Ios	VI	S	-130 0	-20 0	-130 0	-20 0	-130 0	-20 0	mA
E79 E82	Ios	VI	S	-150 0	-60 0	-150 0	-60 0	-150 0	-60 0	mA
E83	Ios	VI	S	-130 0	-20 0	-130 0	-20 0	-130 0	-20 0	mA
E84 E88	Ios	VI	S	-150 0	-60 0	150 0	-60 0	-150 0	-60 0	mA
E89	Ios	VI	S	-130 0	-20 0	-130 0	-20 0	-130.0	-20 0	mA
E90-E94	Ios	VI	S	-150 0	-60 0	-150 0	-60 0	-150.0	-60 0	mA
E95	Ios	VI	S	-130 0	-20 0	-130 0	-20 0	-130 0	-20 0	mA
E96-E98	Ios	VI	S	-150 0	-60 0	-150 0	-60 0	-150 0	-60 0	mA
E99	Ios	VI	S	-130 0	-20 0	-130 0	-20 0	-130 0	-20 0	mA
E100-E103	Ios	VI	S	-150 0	-60 0	-150 0	-60 0	-150 0	-60 0	mA
E104	Ios	VI	S	-130 0	-20 0	-130 0	-20 0	-130.0	-20 0	mA
E105,E106	Ios	VI	S	-150 0	-60 0	-150 0	-60 0	-150 0	-60 0	mA
E107	Ios	VI	S	-130 0	-20 0	-130 0	-20 0	-130 0	-20 0	mA
E108-E113	Ios	VI	S	-150 0	-60 0	-150 0	-60 0	-150 0	-60 0	mA
E114	Ios	VI	S	-130 0	-20 0	-130 0	-20 0	-130.0	-20.0	mA
E115-E119	Ios	VI	S	-150 0	-60.0	-150 0	-60 0	-150 0	-60 0	mA

See footnotes at end of table

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:15Z
Check the source to verify that this is the current version before use.

TABLE I Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/ - 3/	25° ± 5°C 4/		Class I 4/ 0° ± 5°C 60° ± 5°C		Class II 4/ -55° ± 5°C 85° ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
E120	Ios	VI	S	-130 0	-20 0	-130 0	20 0	-130 0	-20.0	mA
E121-E125	Ios	VI	S	-150 0	-60 0	-150 0	-60 0	-150 0	-60 0	mA
E126	Ios	VI	S	-130 0	-20.0	-130.0	-20 0	-130.0	-20 0	mA
E127-E132	Ios	VI	S	-150 0	-60 0	-150 0	60 0	-150 0	-60 0	mA
E133	Ios	VI	S	-130 0	-20 0	-130 0	-20 0	-130 0	-20 0	mA
E134-E138	Ios	VI	S	-150 0	-60 0	-150 0	-60 0	-150 0	-60 0	mA
E139	Ios	VI	S	-130 0	-20 0	130 0	-20 0	-130 0	-20 0	mA
E140 E144	Ios	VI	S	-150 0	-60 0	150 0	60 0	-150 0	-60 0	mA
E145	Ios	VI	S	-130 0	-20 0	-130 0	20 0	-130 0	-20 0	mA
F1-F4	Iih	VII	X	-2 41	0 34	2 41	0 34	-2.41	0 34	mA
F5,F6	Iih	VII	X	-1 71	0 24	-1.71	0 24	-1 71	0.24	mA
F7	Iih	VII	X	-2 41	0 34	-2 41	0 34	-2.41	0.34	mA
F8,F9	Iih	VII	X	-1 71	0 24	1.71	0.24	-1.71	0 24	mA
F10	Iih	VII	X	-2 41	0 34	-2 41	0 34	-2 41	0 34	mA
F11-F22	Iih	VII	X	-0 6	0 02	-0 6	0 02	-0 6	0 02	mA
F23	Iih	VII	X	-1 81	0 32	-1.81	0 32	-1.81	0 32	mA
F24-F26	Iih	VII	X	-1 87	0 37	-1 87	0 37	-1 87	0 37	mA
F27	Iih	VII	X	-1 71	0 24	-1 71	0 24	-1.71	0 24	mA
F28	Iih	VII	X	-2 41	0 34	-2 41	0 34	-2.41	0 34	mA
F29	Iih	VII	X	-1 71	0 24	-1 71	0 24	-1.71	0 24	mA
F30,F31	Iih	VII	X	-27 7	-14 3	-27 7	-14 3	-27 7	-14 3	mA
F32	Iih	VII	X	-0 6	0 02	-0 6	0 02	-0 6	0 02	mA
F33	Iih	VII	X	-1 24	0 08	-1 24	0.08	-1.24	0 08	mA
F34	Iih	VII	X	-4 8	0 08	-4 8	0 08	-4 8	0 08	mA
F35	Iih	VII	X	3 6	0 12	-3 6	0 12	-3 6	0 12	mA
F36 F37	Iih	VII	X	-1 2	0 04	-1 2	0 04	-1 2	0 04	mA
F38	Iih	VII	X	1 8	0 06	-1 8	0.06	-1 8	0 06	mA
F39	Iih	VII	X	-1 2	0 04	-1 2	0 04	-1 2	0 04	mA
F40	Iih	VII	X	27 7	-14 3	-27.7	-14 3	-27.7	-14 3	mA
F41 F44	Iih	VII	X	-0 6	0 02	-0 6	0 02	-0 6	0 02	mA
F45	Iih	VII	X	27 7	-14 3	-27 7	-14 3	-27 7	-14 3	mA
F46 F49	Iih	VII	X	0 6	0 02	-0 6	0 02	-0 6	0 02	mA
F50	Iih	VII	X	-1 27	0 35	-1.27	0 35	-1 27	0 35	mA

See footnotes at end of table

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:15Z
Check the source to verify that this is the current version before use.

TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25° ± 5°C Initial limits Min	Max	Units	Notes
E21	MEM DATA 0	12	0.0	-150.0	-60.0	mA	1-4
E22	MEM DATA 0			-150.0	-60.0		
E23	MEM DATA 0	12		-130.0	-20.0		
E24	MEM DATA 6	13		-150.0	-60.0		
E25	MEM DATA 6						
E26	MEM DATA 6						
E27	MEM DATA 6			-150.0	-60.0		
E28	MEM DATA 6	13		-130.0	-20.0		
E29	COM BUS R6	14		-350.0	-60.0		
E30	COM BUS R6	14					
E31	COM BUS R5	15					
E32	COM BUS R5	15					
E33	COM BUS R4	16					
E34	COM BUS R4	16		-350.0			
E35	MEM DATA 2	17		-150.0			
E36	MEM DATA 2						
E37	MEM DATA 2						
E38	MEM DATA 2			-150.0	-60.0		
E39	MEM DATA 2	17		-130.0	-20.0		
E40	MEM DATA 3	18	0.0	-150.0	-60.0	mA	1-4

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:15Z
Check the source to verify that this is the current version before use.

TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25° ± 5°C Initial limits Min	Max	Units	Notes
E41	MEM DATA 3	18	0.0	-150.0	-60.0	mA	1-4
E42	MEM DATA 3						
E43	MEM DATA 3			-150.0	-60.0		
E44	MEM DATA 3	18		-130.0	-20.0		
E45	COM BUS R3	19		-350.0	-60.0		
E46	COM BUS R3	19					
E47	COM BUS R2	20					
E48	COM BUS R2	20					
E49	COM BUS R1	21					
E50	COM BUS R1	21					
E51	COM BUS R0	22					
E52	COM BUS R0	22		-350.0			
E53	MEM DATA 21	23		-150.0			
E54	MEM DATA 21			-150.0	-60.0		
E55	MEM DATA 21	23		-130.0	-20.0		
E56	MEM DATA 16	24		-150.0	-60.0		
E57	MEM DATA 16			-150.0	-60.0		
E58	MEM DATA 16	24		-130.0	-20.0		
E59	MEM DATA 4	26		-150.0	-60.0		
E60	MEM DATA 4	26	0.0	-150.0	-60.0	mA	1-4

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:15Z
Check the source to verify that this is the current version before use.

TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25° ± 5°C Initial limits Min	Max	Units	Notes
E61	MEM DATA 4	26	0.0	-150.0	-60.0	mA	1-4
E62	MEM DATA 4			-150.0	-60.0		
E63	MEM DATA 4	26		-130.0	-20.0		
E64	MEM DATA 18	27		-150.0	-60.0		
E65	MEM DATA 18			-150.0	-60.0		
E66	MEM DATA 18	27		-130.0	-20.0		
E67	MEM DATA 19	28		-150.0	-60.0		
E68	MEM DATA 19			-150.0	-60.0		
E69	MEM DATA 19	28		-130.0	-20.0		
E70	MEM DATA 1	29		-150.0	-60.0		
E71	MEM DATA 1						
E72	MEM DATA 1						
E73	MEM DATA 1			-150.0	-60.0		
E74	MEM DATA 1	29		-130.0	-20.0		
E75	MEM DATA 7	44		-150.0	-60.0		
E76	MEM DATA 7						
E77	MEM DATA 7			-150.0	-60.0		
E78	MEM DATA 7	44		-130.0	-20.0		
E79	MEM DATA 9	45		-150.0	-60.0		
E80	MEM DATA 9	45	0.0	-150.0	-60.0	mA	1-4

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:15Z
Check the source to verify that this is the current version before use.

TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25° ± 5°C Initial limits Min	Max	Units	Notes
E81	MEM DATA 9	45	0.0	-150.0	-60.0	mA	1-4
E82	MEM DATA 9			-150.0	-60.0		
E83	MEM DATA 9			-130.0	-20.0		
E84	MEM DATA 9	45		-150.0	-60.0		
E85	MEM DATA 10	46					
E86	MEM DATA 10						
E87	MEM DATA 10						
E88	MEM DATA 10			-150.0	-60.0		
E89	MEM DATA 10			-130.0	-20.0		
E90	MEM DATA 10	46		-150.0	-60.0		
E91	MEM DATA 8	47					
E92	MEM DATA 8						
E93	MEM DATA 8						
E94	MEM DATA 8			-150.0	-60.0		
E95	MEM DATA 8			-130.0	-20.0		
E96	MEM DATA 8	47		-150.0	-60.0		
E97	MEM DATA 20	48					
E98	MEM DATA 20			-150.0	-60.0		
E99	MEM DATA 20	48		-130.0	-20.0		
E100	MEM DATA 5	49	0.0	-150.0	-60.0	mA	1-4

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:15Z
Check the source to verify that this is the current version before use.

TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25° ± 5°C Initial limits Min	Max	Units	Notes
E101	MEM DATA 5	49	0.0	-150.0	-60.0	mA	1-4
E102	MEM DATA 5						
E103	MEM DATA 5			-150.0	-60.0		
E104	MEM DATA 5	49		-130.0	-20.0		
E105	MEM DATA 17	50		-150.0	-60.0		
E106	MEM DATA 17			-150.0	-60.0		
E107	MEM DATA 17	50		-130.0	-20.0		
E108	LATCH STATUS	59		-150.0	-60.0		
E109	ENBL COM BUS	61					
E110	MEM DATA 15	76					
E111	MEM DATA 15						
E112	MEM DATA 15						
E113	MEM DATA 15			-150.0	-60.0		
E114	MEM DATA 15			-130.0	-20.0		
E115	MEM DATA 15	76		-150.0	-60.0		
E116	MEM DATA 13	78					
E117	MEM DATA 13						
E118	MEM DATA 13						
E119	MEM DATA 13			-150.0	-60.0		
E120	MEM DATA 13	78	0.0	-130.0	-20.0	mA	1-4

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:15Z
Check the source to verify that this is the current version before use.

TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25° ± 5°C Initial limits Min	Max	Units	Notes
E121	MEM DATA 13	78	0.0	-150.0	-60.0	mA	1-4
E122	MEM DATA 14	79					
E123	MEM DATA 14						
E124	MEM DATA 14						
E125	MEM DATA 14			-150.0	-60.0		
E126	MEM DATA 14			-130.0	-20.0		
E127	MEM DATA 14	79		-150.0	-60.0		
E128	WR DATA 15	83					
E129	MEM DATA 12	87					
E130	MEM DATA 12						
E131	MEM DATA 12						
E132	MEM DATA 12			-150.0	-60.0		
E133	MEM DATA 12			-130.0	-20.0		
E134	MEM DATA 12	87		-150.0	-60.0		
E135	MEM DATA 11	88					
E136	MEM DATA 11						
E137	MEM DATA 11						
E138	MEM DATA 11			-150.0	-60.0		
E139	MEM DATA 11			-130.0	-20.0		
E140	MEM DATA 11	88	0.0	-150.0	-60.0	mA	1-4

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:15Z
Check the source to verify that this is the current version before use.

TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25° ± 5°C Initial limits Min	Max	Units	Notes
E141	WR DATA 14	91	0.0	-150.0	-60.0	mA	1-4
E142	LPBCK SEL2	96					
E143	SOFT ERROR	98					
E144	BIT ERROR	99		-150.0	-60.0		
E145	ECC ERROR	100	0.0	-130.0	-20.0	mA	1-4

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:15Z
Check the source to verify that this is the current version before use.

NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	E1-E145
Power supply current limit	2.52 A	NA	E1-E145
Forcing voltage, (Vout)	0.0 V	± 50 mV	E1-E145
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 E1-E145 E1-E145
Output current measurement	NA

NOTES: - Continued.

Condition	Value	Accuracy	Test condition
	NA	± 5.5 mA	E74
		± 4.5 mA	E75-E77
		± 5.5 mA	E78
		± 4.5 mA	E79-E82
		± 5.5 mA	E83
		± 4.5 mA	E84-E88
		± 5.5 mA	E89
		± 4.5 mA	E90-E94
		± 5.5 mA	E95
		± 4.5 mA	E96-E98
		± 5.5 mA	E99
		± 4.5 mA	E100-E103
		± 5.5 mA	E104
		± 4.5 mA	E105,E106
		± 5.5 mA	E107
		± 4.5 mA	E108-E113
		± 5.5 mA	E114
		± 4.5 mA	E115-E119
		± 5.5 mA	E120
		± 4.5 mA	E121-E125
		± 5.5 mA	E126
		± 4.5 mA	E127-E132
		± 5.5 mA	E133
		± 4.5 mA	E134-E138
		± 5.5 mA	E139
		± 4.5 mA	E140-E144
	NA	± 5.5 mA	E145

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:15Z
Check the source to verify that this is the current version before use.

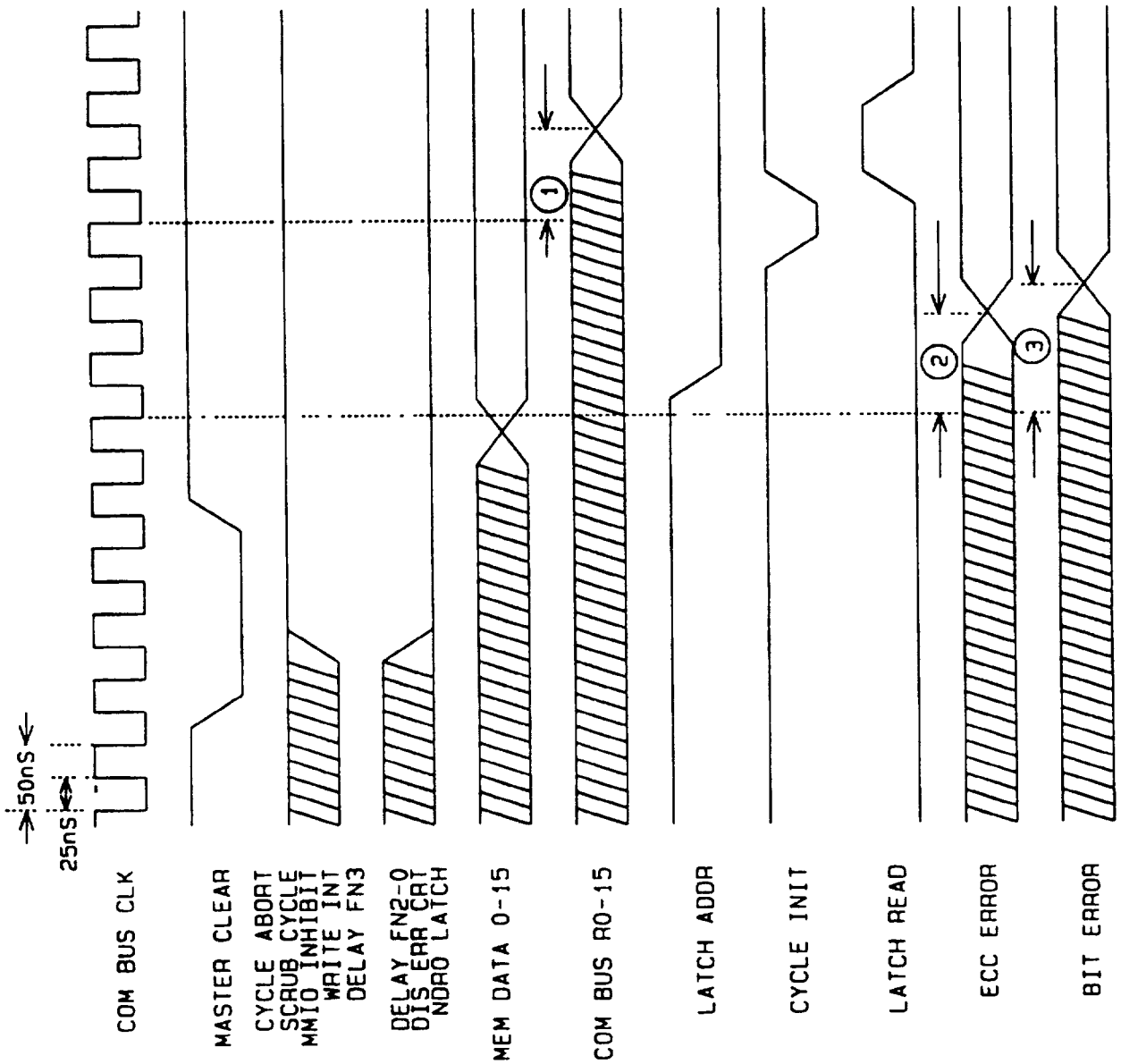


FIGURE 12a. Dynamic waveforms for table XVII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:15Z
Check the source to verify that this is the current version before use.